



Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Line Coding Techniques
Lab Number	09
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

LAB: 09

Topic: Line Coding Techniques

Objective:

The objective of this lab is implementation of line coding techniques using Mathscript LabVIEW.

Introduction:

A line code is the code used for data transmission of a digital signal over a transmission line.

Types of Line Coding:

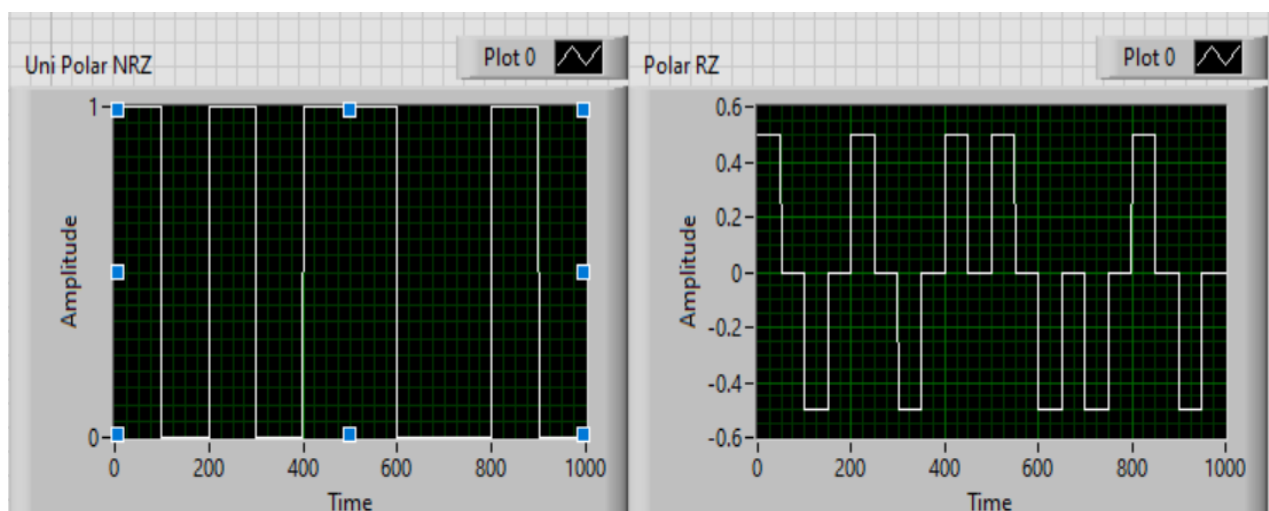
There are 3 types of Line Coding

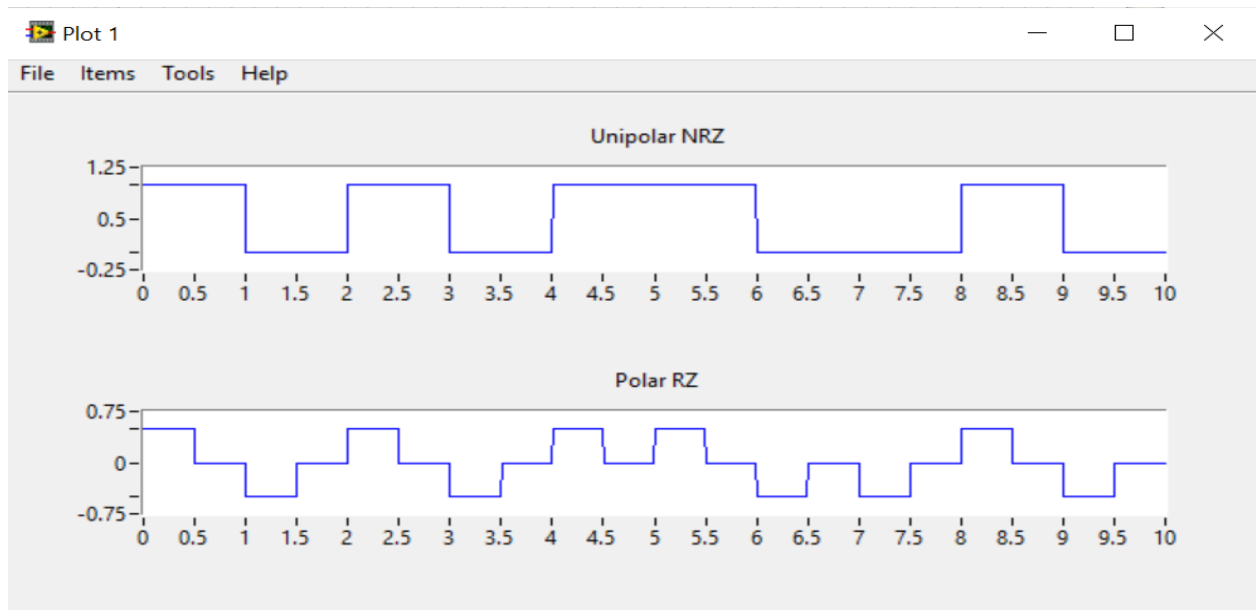
1. Unipolar.
2. Polar.
3. Bi-polar.

Task 1:

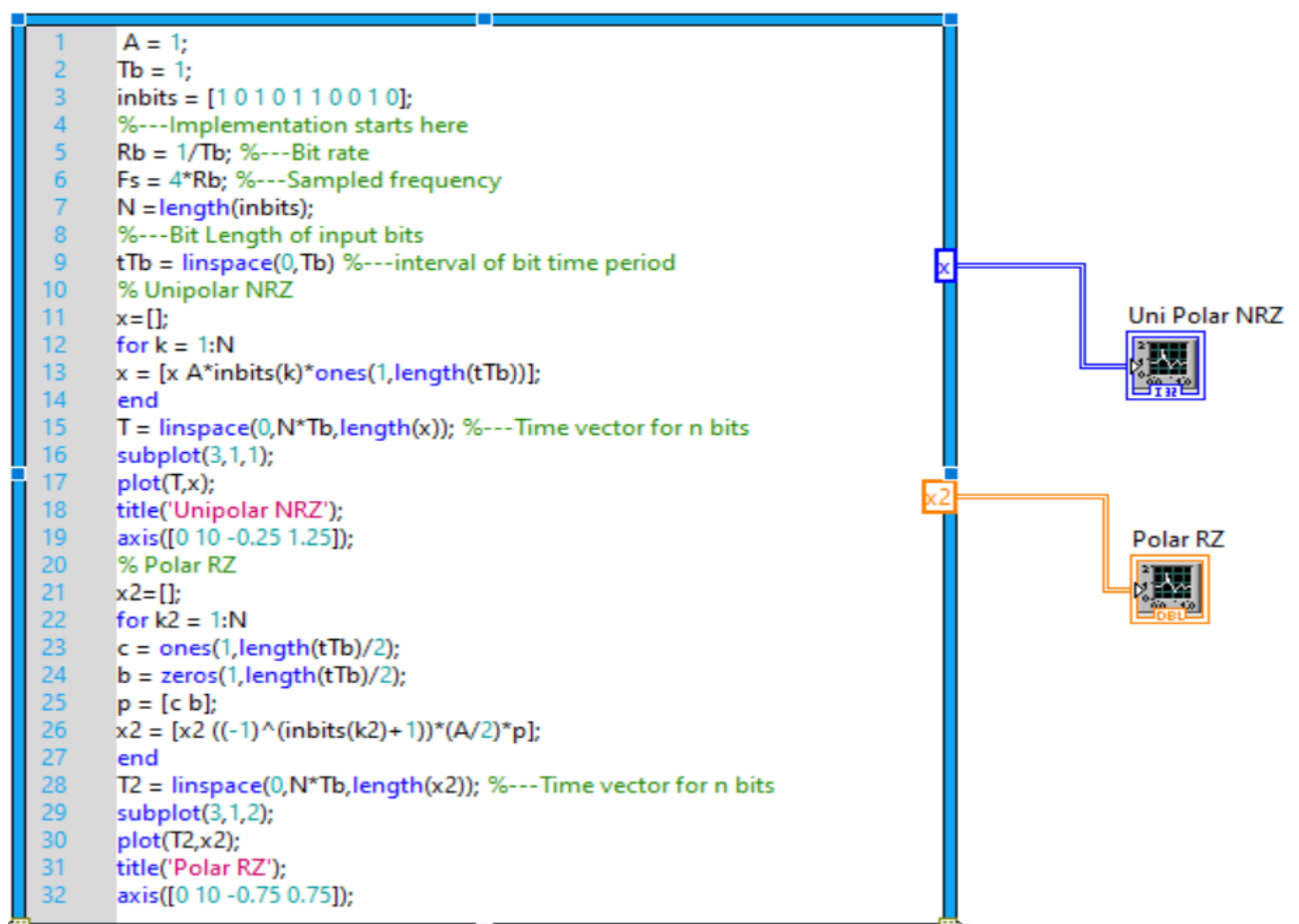
Implement Unipolar NRZ, Polar RZ and Polar NRZ-L for bit stream [1 0 1 0 1 1 0 0 1 0].

FRONT PANEL:



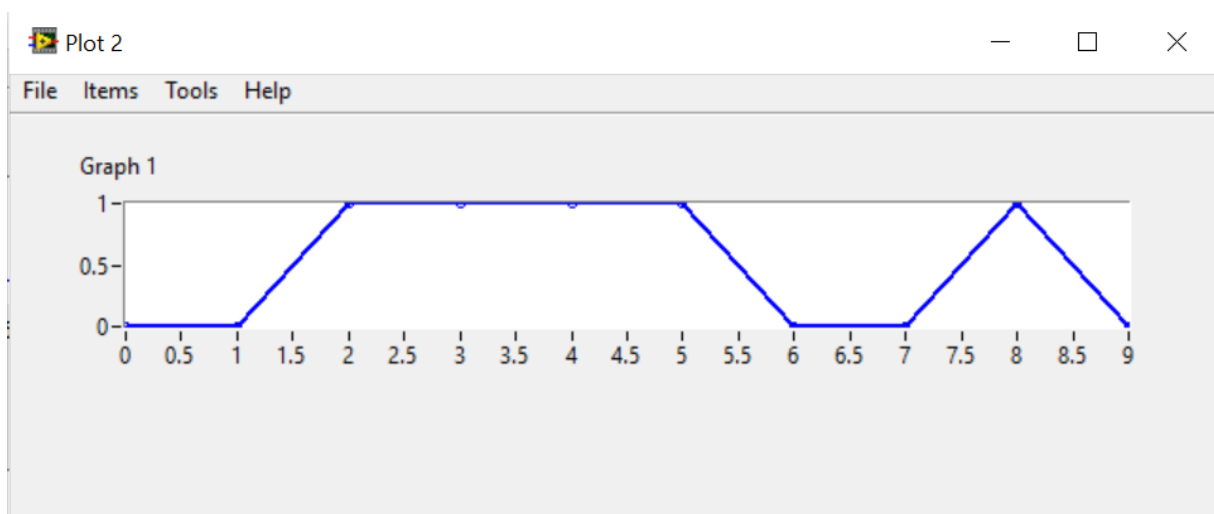
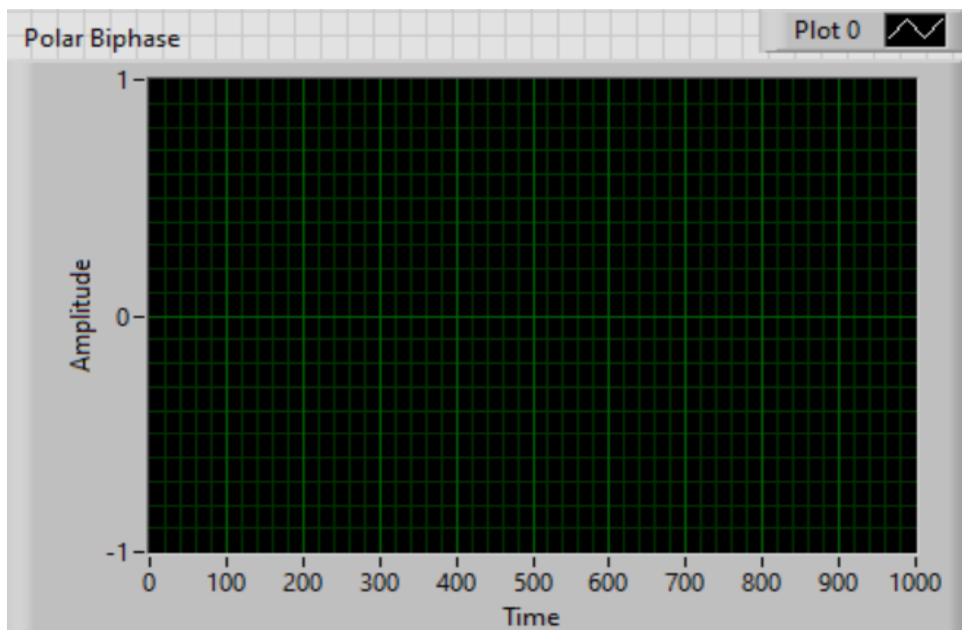


Block Diagram:



Task 2:

Implement Polar Biphase Manchester for bit stream [0 0 1 1 1 1 0 0 1 0].



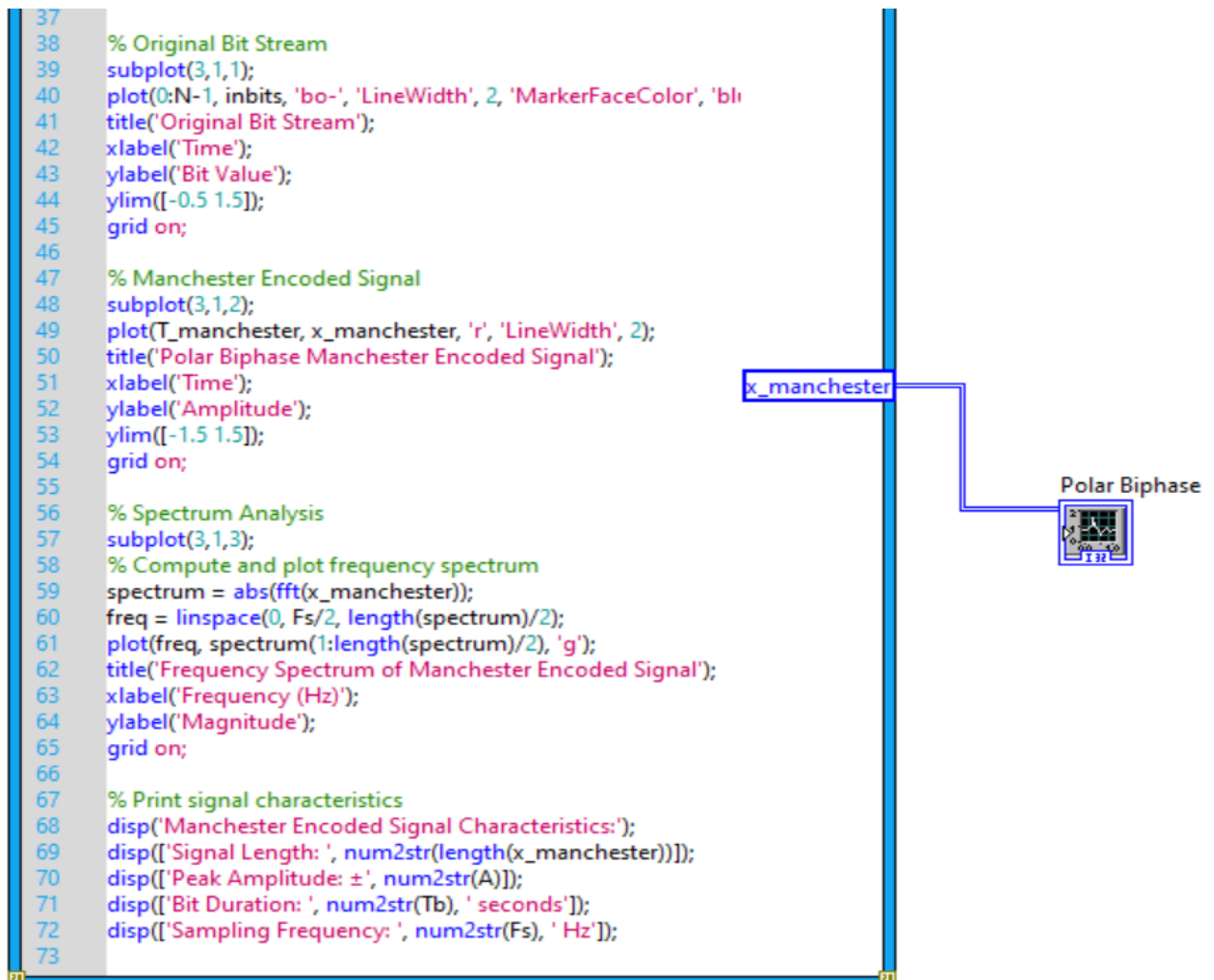
Block Diagram:

```
1 % Input bit stream
2 inbits = [0 0 1 1 1 1 0 0 1 0];
3
4 % Encoding Parameters
5 A = 1; % Amplitude
6 Tb = 1; % Bit time period
7 Rb = 1/Tb; % Bit rate
8 Fs = 4*Rb; % Sampled frequency
9 N = length(inbits); % Number of input bits
10
11 % Time vector for bit period
12 tTb = linspace(0, Tb, Fs);
13
14 % Manchester Encoding
15 x_manchester = [];
16 for k = 1:N
17     % Manchester Encoding Logic
18     if inbits(k) == 0
19         % 0: Low to High transition
20         first_half = -A * ones(1, length(tTb)/2);
21         second_half = A * ones(1, length(tTb)/2);
22     else
23         % 1: High to Low transition
24         first_half = A * ones(1, length(tTb)/2);
25         second_half = -A * ones(1, length(tTb)/2);
26     end
27
28     % Combine first and second half
29     x_manchester = [x_manchester, first_half, second_half];
30 end
31
32 % Time vector for Manchester encoded signal
33 T_manchester = linspace(0, N*Tb, length(x_manchester));
34
35 % Plotting
36 figure;
```

x_manchester

Polar Biphase





Conclusion:

This lab focused on the implementation of line coding techniques using Mathscript in LabVIEW. Line coding is essential for transmitting digital signals over a transmission line, and we explored three key types: Unipolar, Polar, and Bipolar. Through this lab, we gained practical experience in encoding and decoding digital data signals, understanding how different line coding methods affect signal representation. The exercise helped demonstrate the importance of selecting appropriate line coding schemes for reliable and efficient data transmission in digital communication systems.